

# PATRICK ADRIANUS

Melbourne, Australia | [patrickadrianus.com](https://patrickadrianus.com)  
<https://www.linkedin.com/in/patrick-adrianus/>  
0421806625 | patrickadrianus04@gmail.com

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Data Science professional with a Bachelor of Data Science (RMIT, 2025), skilled in Python, R, and SQL for predictive analytics, machine learning, and data visualization. Experienced in building end-to-end data pipelines and analytical dashboards that deliver actionable insights. Passionate about applying ML and automation to support smarter business decisions and innovation. Currently working at Woolworths Group in store operations and compliance, where I combine technical and analytical skills to improve data reliability, process efficiency, and team performance.

Technical focus: Machine Learning, Data Engineering, and Applied Analytics using Python, SQL, and cloud-based tools.

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## EXPERIENCE

### SOFTWARE DEVELOPMENT INTERNSHIP

Urban Waste, Victorian Government | Melbourne, Australia

Dec 2022 – Feb 2023

- Supported ERP system design for data integration and reliability across multiple departments.
  - Collaborated with cross-functional teams to resolve operational issues and streamline workflows.
  - Developed and maintained Python/XML/HTML modules to enhance backend usability.
  - Contributed to Agile sprints and stakeholder updates to ensure project alignment.
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## PROJECTS

### LLM-BASED DETECTION OF PRIVACY ISSUES IN STACK OVERFLOW (CAPSTONE PROJECT, 2025)

- Collected and filtered 10,000+ Stack Overflow posts via API, then manually labeled 1,000+ entries for supervised fine-tuning and evaluation.
- Tested and benchmarked 5+ model-prompt configurations (GPT, DeepSeek, Mistral). Found DeepSeek Few-Shot prompting achieved the best recall (~48%) for privacy-violation detection.
- Designed a multi-stage pipeline (keyword filtering > manual labeling > LLM classification) achieving precision of 0.61 and F1 of 0.54, outperforming baseline heuristics.
- Coordinated 4 two-week Agile sprints with JIRA tracking, maintaining 100% on-time delivery and seamless communication with supervisor and client.
- Proposed improvements including hybrid rules + LLM filtering and fine-tuned transformer training for future scalability.

### CLINIWATCH (RMIT HACKATHON, 2025)

- Built an iOS prototype for early mental-health detection integrating Apple HealthKit, SwiftUI, and Core ML visualization APIs.
- Enabled real-time tracking of 5+ metrics (mood, sleep, energy, activity, journaling) with automated alert flags for carers and clinicians.
- Implemented a dashboard with streaks, aura points, and motivational feedback, improving user engagement by an estimated 40% during pilot demo.
- Collaborated with a 5-member multidisciplinary team, delivered a working prototype in under 48 hours, earning a Special Mention for Innovation and User Impact from Blue & Northern Health.

### HUMAN ACTION RECOGNITION (RMIT, 2025)

- Designed and trained a CNN architecture (3D ConvNet) to classify 9 human actions using a dataset of 4,500 images, augmented to 18,000 frames.
  - Tackled class imbalance with focal loss + dynamic weighting, increasing minority-class recall by 22% compared to baseline BCE.
  - Achieved overall accuracy = 87.3%, outperforming initial model by >12 points.
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- Benchmarked several architectures and produced a reproducible Jupyter Notebook pipeline with documented preprocessing and evaluation.

### AI-POWERED FAKE NEWS DETECTION (RMIT, 2023)

- Conducted comparative analysis of Random Forest vs SVM for fake-news classification using a corpus of 2,000 articles.
- Optimized feature extraction with TF-IDF, achieving 91% accuracy and 0.88 F1 score with Random Forest.
- Evaluated algorithmic fairness through bias checks and three independent data audits ensuring ethical AI outcomes.
- Delivered a group presentation to 50+ attendees, linking model interpretability to journalistic transparency and governance practices.

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## OTHER EXPERIENCE

### FRESH CONVENIENCE & STORE SERVICES ASSISTANT (SSA)

Woolworths | Melbourne, Australia

Mar 2023 – Present

- Maintain product quality, stock rotation, and replenishment to meet company standards.
- Deliver excellent customer service in a fast-paced retail environment.
- Support payroll performance and cash handling accuracy across assisted and self-serve checkouts.
- Verify daily cash flow through compliance checks to maintain operational accuracy.
- Collaborate with store management to improve efficiency and customer experience.

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## EDUCATION

### IT PROFESSIONAL YEAR PROGRAM

ECA, Melbourne

Sept 2025 – Present (Ongoing)

### BACHELOR OF DATA SCIENCE

RMIT University, Melbourne

July 2022 - June 2025

- Graduated with Distinction (GPA 3.5/4.0).
- Coursework: Machine Learning, Deep Learning, Big Data, Multivariate Analysis, Cybersecurity.

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## ACHIEVEMENTS

Treasurer, PPIA RMIT (2023–2024); Event Organizer, PPIA RMIT (2022–2023); Participant, Apple Foundation Program (2024); Participant, RMIT Hackathon (2025).

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## SKILLS

### TECHNICAL SKILLS:

- **Data Science & ML:** Regression, Classification, Clustering, Deep Learning, NLP, Data Visualization
- **Programming:** Python, R, Swift, SQL, JavaScript, HTML/CSS, XML
- **Frameworks/Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Flask, React, Vite, spaCy, Seaborn, Matplotlib
- **Platforms & Tools:** Git, Jupyter, Anaconda, RStudio, Power BI, AWS (S3, Lambda), Xcode (iOS Dev)

### INTERPERSONAL SKILLS:

- Clear communication of technical insights
  - Problem solving & critical thinking
  - Attention to detail
  - Collaboration in Agile/Scrum teams
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- Mentoring & teamwork
  - Time management under deadlines
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